FOSSBYTES

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Apple open sources kernel of iOS and macOS for ARM processors

Apple has open sourced the kernel code of the latest macOS X and iOS updates. The company always shares the kernels of macOS and iOS on GitHub, but it has this time enabled the experience for ARM chips.

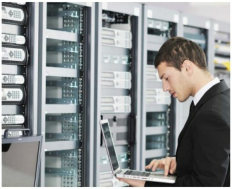
One of the reasons why Apple open sources a portion of macOS and iOS each year is because of its indirect dependence on the BSD project. There is no way to let developers use the open source codebase published by Apple to compile their own OS. However, the kernel guides coders to begin with macOS and iOS kernels.

Notably, the ARM-based source code of Apple's kernel doesn't mean much but gives Apple a way to give back to the open source community. Some open source experts are predicting that the iPhone maker could be working on a version of macOS that runs on ARM chips.

Oracle develops Fn to enter serverless computing world

Oracle is all set to step into the nascent space of serverless computing by developing an all-new open source, cloud-agnostic platform called Fn. The new open source development can be run either on-premise or in the cloud and works with Docker containers.

"The Fn project is a container-native Apache 2.0 licensed serverless platform that you can run anywhere — on any cloud or on-premise," said Shaun Smith, product management director at Oracle, in a blog post.



Oracle's Fn has a command line interface that helps developers build, test and deploy functions using orchestration tools such as Kubernetes, Mesosphere and Docker Swarm. There is a built-in smart load balancer that routes traffic directly to functions to enhance the experience.

You can use Fn with functions written in Java at the initial stage. But there are also plans to expand the platform by adding support for Go, Ruby, Python, PHP and Node.js over time.

Along with the other features, the Fn project allows developers to take AWS Lambda functions and run them platform agnostically. You need to use the *lambda-node-4* runtime to begin with Lambda functions on Fn.

Kotlin 1.2 beta comes with support for Java 9

Kotlin 1.2 beta has arrived. With the new release, developers can build some interesting modern multi-platform applications. The experimental multi-platform

Google launches Cloud Firestore to help developers store app data easily

To enhance developer engagements around its Firebase application development platform, Google has launched Cloud Firestore. The latest innovation brings a fully-managed NoSQL document database that can store and sync app data at a global scale.

"We've aimed for the simplicity and ease-of-use that is always top priority for Firebase, while making sure that Cloud Firestore can scale to power even the largest apps," said Alex Dufetel, product manager for Firebase at Google, in a blog post.

Available in beta, Cloud Firestore is designed to help developers hold documents and collections on the cloud with querying support. The database comes with iOS, Android and Web SDKs. It also supports offline data access and enables real-time data synchronisation. Additionally, developers can leverage the built-in automatic, multi-region data replication feature to get better cloud access.

The client-side SDKs of Cloud Firestore enable serverless development and provide offline data access with local database support for the Web as well as Android and iOS platforms. Developers can pick the Cloud Firestore SDKs for Java, Go, Python and Node.js.

Google's Cloud Firestore hasn't been developed from scratch, but built as an upgrade to the existing Firebase Realtime Database, which offers the developer community a real-time database — managed and scaled by Google. Key limitations such as data structuring, querying and scaling have all been addressed in this upgrade.

You can get started with the public beta of Cloud Firestore by visiting the database tab in your Firebase console.